# Junwei Huang

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Research Interests: Power management ICs (PMICs); high current density; fast transient response; chiplet/vertical power delivery

## EDUCATION

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| **2018.08–2024.07** | University of Macau | Electrical and Computer Engineering | Ph.D.<br>Advisors: Prof. Yan Lu and Prof. Chi-Seng Lam |
| **2014.09–2018.06** | University of Electronic Science and Technology of China | Microelectronics Science and Engineering | B.Eng. |

## ACADEMIC APPOINTMENTS

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| **2025.09–present** | University of California, Berkeley | Department of Electrical Engineering and Computer Sciences (EECS) | Visiting Postdoctoral Researcher<br>Faculty Host: Prof. Robert Pilawa-Podgurski |
| **2025.09–present** | University of Macau | Institute of Microelectronics | Postdoctoral Fellow<br>PI: Prof. Rui P. Martins and Prof. Sai-Weng Sin |
| **2024.11–2025.05** | Tsinghua University | Department of Electronic Engineering | Visiting Scholar<br>Faculty Host: Prof. Yan Lu |
| **2024.10–2025.09** | University of Macau | Institute of Microelectronics | Research Assistant<br>PI: Prof. Rui. P Martins |

## RESEARCH SUMMARY

**University of Macau, State Key Laboratory of Analog and Mixed-Signal VLSI** **2018.08–present**

- My research focuses on the design of power management chips with high current density and fast transient response. To date, I have published 25 papers in international conferences and journals, including 17 conference papers at ISSCC (1 highlight paper as first author), CICC (3 paper as first author), and ISCAS, and 8 journals in JSSC (2 as first author), TCAS-I (1 as first author), OJ-SSCS, and JOS
- I was invited to present at an IEEE PES/PELS event held at UC Berkeley, recognized for expertise in PMIC design.

**University of California, Berkeley, EECS** **2025.09–present**

- My research focuses on the co-design of board-level power electronics and on-package/on-die PMICs, targeting >3,000 A aggregate load and >2.5 A/mm² current density and >4550W/in³ power density for next-generation CPUs/GPUs.

## AWARDS AND ACADEMIC SERVICE

1. Akrostar Technology Academic Prize for the academic year 2023/2024 (Top 3).
2. Reviewer for IEEE Transactions on Industrial Electronics, IEEE Transactions on Circuits and Systems II.

## RESEARCH WORK AND PUBLICATIONS

**Conference:**

[C1] **J. Huang**, X. Mao, Z. Tong, Z. Yu, W. Yang, C. -S. Lam, R. P. Martins, Y. Lu, "A 20MHz-1MHz Dual-Loop Non-Uniform-Multi-Inductor Hybrid DC-DC Converter with Specified Inductor Current Allocation and Fast Transient Response," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2025. **Highlight Paper**

[C2] **J. Huang**, Z. Tong, X. Mao, C. -S. Lam, R. P. Martins and Y. Lu, "A Fast-Slow Two-Module DC-DC Solution with Transient and Efficiency Improvements for 2.5D/3D Integration," 2024 IEEE Custom Integrated Circuits Conference (CICC), Apr. 2024.

[C3] **J. Huang**, Z. Tong, Y. Lu, C. -S. Lam, R. P. Martins, "A 5V-to-0.5V Inductor-First Inductor-on-Ground Switched Capacitor Multi-Path Hybrid DC-DC Converter," 2023 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2023.

[C4] **J. Huang**, X. Mao, Z. Tong, Z. Yu, S. Han, C. -S. Lam, R. P. Martins, Y. Lu, "A Scalable Distributed 12-to-1V Fast-Slow Hybrid DC-DC Combo with Fast-Transient Bucklets and One-Pin Current Balancing," 2026 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2026. **Accepted**

[C5] Z. Tong, **J. Huang**, Y. Lu, R. P. Martins, "A 42W Reconfigurable Bidirectional Power Delivery Voltage-Regulating Cable," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2023. **Highlight Paper**

[C6] Z. Zhu, **J. Huang**, Z. Song, Z. Yu, S. Han, S-W. Sin, Y. Lu "An Inductor-at-Middle Hybrid Buck Converter with Shared Power-Signal Path for Distributed Vertical Power Delivery," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2026. **Accepted**

[C7] Z. Yu, Z. Tong, **J. Huang**, J. Yin, Y. Lu "A Compact Dual-Capacitor Relay SPT Supply Modulator with Overshoot-Free Adaptive On-Time Control for 5G FR2 CMOS PA," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2026. **Accepted**

[C8] Z. Tong, Z. Yu, **J. Huang**, X. Mao, Bernhard Wicht, Rui P. Martins, Yan Lu, "HOOP: A Scalable Hybrid DC-DC Converter Ring for High Performance Computing," in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2025.

[C9] W. Yang, Z. Tong, **J. Huang**, R. P. Martins, Y. Lu, "A Bi-Directional Dual-Path Boost-48V-Buck Hybrid Converter for High-Voltage Power Transmission Cable in Light-Weight Humanoid Robots" in IEEE Int. Solid-State Circuits Conf. (ISSCC), Feb 2025.

[C10] Z. Yu, **J. Huang**, Z. Tong, Z. Zhu, Y. Lu, "An Output-Rail-Scalable Hybrid Buck Converter with Globally Adaptive Frequency Control Achieving 98.0% Peak Efficiency and 490 mW/mm³ Power Density," 2026 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2026. **Accepted**

[C11] S. Han, **J. Huang**, F. Song, Z. Tong, S-W Sin, H. Jiang, X. Wu, T. Ren, Y. Lu, "A 94.1% Peak Efficiency Coupled-Inductor Boost Converter with Small Inductor Current Ripple and Fast Transient Response," 2026 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2026. **Accepted**

[C12] Z. Song, Z. Tong, **J. Huang**, Z. Zhu, S-W Sin, Y. Lu, "A 6V-to-1V ReSC-Hybrid Two-Stage Merged Buck Converter with an Ultra-Small Inductor for VPD," 2026 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2026. **Accepted**

[C13] X. Mao, **J. Huang**, Z. Tong, R. P. Martins and Y. Lu, "A Quad-Output Hybrid Buck Converter with 8-Inductor Helping One Spot from All Quarters for Multi-Core XPUs," 2024 IEEE Custom Integrated Circuits Conf. (CICC), Apr. 2024.

[C14] Z. Tong, W. Yang, S. Han, **J. Huang**, X. Mao, Y. Lu, "Where is the Inductor: A Review and Comparison of the Hybrid DC-DC Buck Topologies," 2025 IEEE Custom Integrated Circuits Conf. (CICC), April 2025.

[C15] Z. Yu, **J. Huang**, Z. Tong, M. Huang and Y. Lu, "An Always Dual-Path Hybrid DC-DC Converter with Multiphase Interleaving Switched-Capacitor Cell Obtaining 45% Output Ripple Reduction," IEEE International Symposium on Circuits and Systems (ISCAS), May 2025.

[C16] F. Luo, **J. Huang**, M. Huang and Y. Lu, "A 12V-Input 1.8V-0.8V-Output Multiple-Output Hybrid Buck DC-DC Converter with a Shared Flying Capacitor," IEEE International Symposium on Circuits and Systems (ISCAS), May 2025.

[C17] Y. Hu, **J. Huang**, M. Huang and Y. Lu, "A 5V-to-0.8V Inductor-First 2L2C Multi-Path Hybrid DC-DC Converter," IEEE International Symposium on Circuits and Systems (ISCAS), May 2025.

**Journal:**

[J1] **J. Huang**, Z. Tong, X. Mao, C. -S. Lam, R. P. Martins and Y. Lu, "A Fast-Slow Two-Module High-Power-Density DC–DC Solution With Transient and Efficiency Improvements," in IEEE Journal of Solid-State Circuit (JSSC), 2025.

[J2] **J. Huang**, Z. Tong, C. -S. Lam, X. Mao, R. P. Martins, Y. Lu, "A Multi-Path Inductor-First Inductor-on-Ground Switched-Capacitor Hybrid DC-DC Converter," in IEEE Journal of Solid-State Circuit (JSSC), 2024.

[J3] **J. Huang**, C. -S. Lam, Y. Lu and R. P. Martins, "A Symmetrical Double Step-Down Converter with Extended Voltage Conversion Ratio," in IEEE Trans. Circuits Syst. I: Regular Papers (TCAS-I), 2022.

[J4] Z. Tong, **J. Huang**, X. Mao, R. P. Martins and Y. Lu, "A Double Pulse Overlapping Laser Diode Driver with Minimum 100-ps Pulse for LiDAR System", in IEEE Journal of the Solid-State Circuits (JSSC), 2024.

[J5] Z. Tong, **J. Huang**, X. Mao, R. P. Martins and Y. Lu, "A Bidirectional USB Power Delivery Voltage-Regulating Cable", in IEEE Journal of the Solid-State Circuits (JSSC), 2024.

[J6] W. Yang, Z. Tong, **J. Huang**, R. P. Martins and Y. Lu, "A Bi-Directional Dual-Path Boost–Buck Hybrid Converter for High-Voltage Power Transmission Delivery Cable in Humanoid Robots," in IEEE Journal of the Solid-State Circuits (JSSC), 2025.

[J7] Y. Lu, **J. Huang**, Z. Tong, T. Hu, W.-L. Zeng, M. Huang, X. Mao and G. Cai, "An Overview of Hybrid DC–DC Converters: From Seeds to Leaves," in IEEE Open Journal of the Solid-State Circuits Society (OJSSC), 2024.

[J8] Y. Lu, G. Cai, and **J. Huang**, Favorable basic cells for hybrid DC–DC converters[J]. Journal of Semiconductors (JOS). 2023.